

# Analysis

Anonymous

2026-09-17

TPOT with bagged individuals -> ensemble

versus

TPOT individual

Version 1

## Stats Testing

```
## # A tibble: 23 x 9
##   task_id .y. group1      group2  n1  n2 statistic      p p.signif
##   <int> <chr> <chr>      <chr> <int> <int>   <dbl>   <dbl> <chr>
## 1   2073 auroc ensemble_random_~ indiv~    7    7     17  6.88e-1 ns
## 2  146818 auroc ensemble_random_~ indiv~   21   21    136  2.55e-1 ns
## 3  146820 auroc ensemble_random_~ indiv~   13   13     43  8.93e-1 ns
## 4  168350 auroc ensemble_random_~ indiv~   17   17     50  2.25e-1 ns
## 5  168757 auroc ensemble_random_~ indiv~   21   21    108.  8.21e-1 ns
## 6  168784 auroc ensemble_random_~ indiv~   20   20     76  2.94e-1 ns
## 7  190137 auroc ensemble_random_~ indiv~   20   20    176  6.39e-3 **
## 8  190146 auroc ensemble_random_~ indiv~   21   21     66  8.88e-2 ns
## 9  190411 auroc ensemble_random_~ indiv~   18   18     48  1.08e-1 ns
## 10 359955 auroc ensemble_random_~ indiv~   21   21    167  7.63e-2 ns
## 11 359956 auroc ensemble_random_~ indiv~   21   21    106  7.59e-1 ns
## 12 359958 auroc ensemble_random_~ indiv~   21   21    206  8.52e-4 ***
## 13 359959 auroc ensemble_random_~ indiv~   19   19     80  5.68e-1 ns
## 14 359960 auroc ensemble_random_~ indiv~   14   14      0  2.25e-2 *
## 15 359962 auroc ensemble_random_~ indiv~   21   21    69.5 1.14e-1 ns
## 16 359963 auroc ensemble_random_~ indiv~   21   21     80  2.29e-1 ns
## 17 359964 auroc ensemble_random_~ indiv~    1    1      1  1 e+0 ns
## 18 359965 auroc ensemble_random_~ indiv~   12   12     47  5.69e-1 ns
## 19 359968 auroc ensemble_random_~ indiv~   17   17     20  5.57e-3 **
## 20 359969 auroc ensemble_random_~ indiv~    1    1      0  1 e+0 ns
## 21 359972 auroc ensemble_random_~ indiv~   20   20    161  3.62e-2 *
## 22 359974 auroc ensemble_random_~ indiv~    7    7      4  1.09e-1 ns
## 23 359975 auroc ensemble_random_~ indiv~    6    6      8  6.88e-1 ns
```

## Medians

```
## # A tibble: 23 x 3
##   task_id individual_tpot ensemble_random_100
##   <int>          <dbl>          <dbl>
## 1   2073          0.871          0.872
## 2  146818          0.932          0.935
## 3  146820          0.998          0.997
## 4  168350          0.964          0.964
## 5  168757          0.790          0.795
## 6  168784          0.967          0.967
## 7  190137          0.918          0.921
## 8  190146          0.963          0.961
## 9  190411          0.921          0.918
## 10 359955          0.743          0.754
## 11 359956          0.932          0.928
## 12 359958          0.934          0.940
## 13 359959          0.751          0.747
## 14 359960          1          1.00
## 15 359962          0.844          0.840
## 16 359963          0.995          0.994
## 17 359964          0.993          0.994
## 18 359965          1.00          1.00
## 19 359968          0.932          0.913
## 20 359969          0.845          0.836
## 21 359972          0.993          0.994
## 22 359974          0.842          0.790
```

|    |    |        |       |       |
|----|----|--------|-------|-------|
| ## | 23 | 359975 | 0.995 | 0.992 |
|----|----|--------|-------|-------|